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CO3-AT1

CSA0708-COMPUTER NETWORKS

Experiments

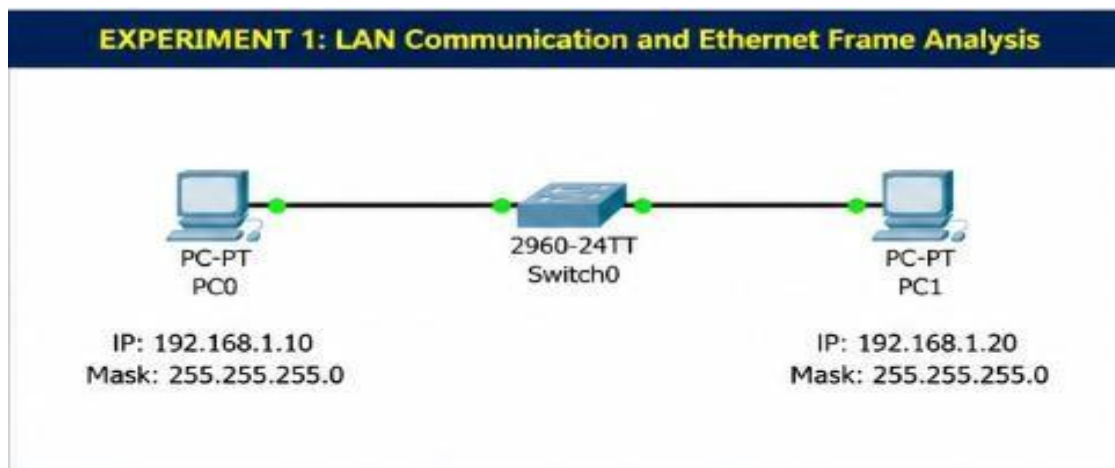
Experiment1:LAN Communication and Ethernet Frame Analysis Aim

To design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer, verify connectivity using the ping command, and analyze Ethernet frame fields using Wireshark.

Requirements

- Cisco Packet Tracer
- Wireshark
- 2 PCs
- 1 Ethernet switch
- Ethernet cables

Network Topology



IP Address Configuration

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.20	255.255.255.0

No default gateway is required because both PCs are in the same LAN.

Procedure

1. Open **Cisco Packet Tracer**.
2. Add two PCs and one switch.
3. Connect PC0 to the switch using a copper straight-through cable.
4. Connect PC1 to the switch.
5. Configure PC0 with:
 - o IP Address: 192.168.1.10
 - o Subnet Mask: 255.255.255.0
6. Configure PC1 with:
 - o IP Address: 192.168.1.20
 - o Subnet Mask: 255.255.255.0
7. Open the Command Prompt of PC0.
8. Execute:

ping 192.168.1.20

Expected Result

Reply from 192.168.1.20

Reply from 192.168.1.20

Reply from 192.168.1.20

Reply from 192.168.1.20

Therefore, communication between the two PCs is successful.

Ethernet Frame Analysis

The important Ethernet frame fields are:

Field	Description
Source MAC Address	MAC address of the sending device
Destination MAC Address	MAC address of the receiving device
EtherType	Identifies the upper-layer protocol
FCS	Used to detect errors in the Ethernet frame

Wireshark Analysis

After capturing the packet:

1. Start Wireshark.
2. Select the active network interface.
3. Start packet capture.
4. Generate traffic using the ping command.
5. Stop the capture.
6. Select an Ethernet frame.
7. Expand the **Ethernet II** section.

The Ethernet frame will contain:

Destination MAC

Source MAC

EtherType

Payload

Frame Check Sequence (FCS)

Role of MAC Address

The Data Link layer uses MAC addresses to deliver frames within the local network. The switch examines the destination MAC address and forwards the Ethernet frame to the appropriate port.

Result

The two PCs successfully communicated through the switch. The Ethernet frame fields were identified using Wireshark, and the role of MAC addressing in local frame delivery was analyzed.

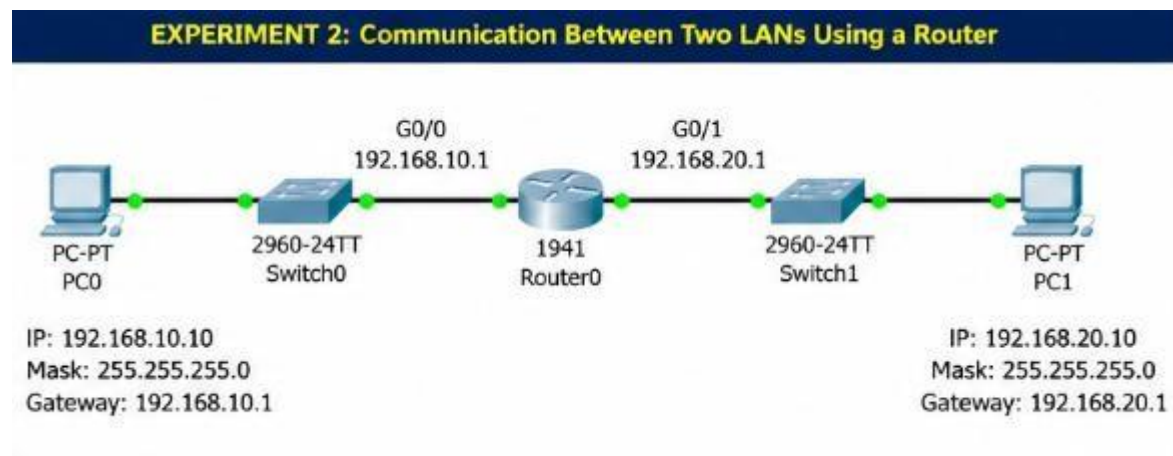
Experiment 2: Communication Between Two LANs Using a Router Aim

To create two LANs connected through a router, configure IPv4 addressing and default gateways, verify communication using ping, and analyze ICMP packets using Wireshark.

Requirements

- CiscoPacketTracer
- Wireshark
- 2 PCs
- 2 switches
- 1 router
- Ethernetcables

NetworkTopology



IPAddressConfiguration

LAN 1

Device	IPAddress	SubnetMask	Gateway
PC0	192.168.10.10	255.255.255.0	192.168.10.1
RouterG0/0	192.168.10.1	255.255.255.0	—

LAN2

Device	IPAddress	SubnetMask	Gateway
RouterG0/1	192.168.20.1	255.255.255.0	—
PC1	192.168.20.10	255.255.255.0	192.168.20.1

RouterConfiguration

enable

configureterminal

interfacegigabitEthernet0/0

```
ipaddress192.168.10.1255.255.255.0
```

```
noshutdown
```

```
exit
```

```
interfacegigabitEthernet0/1
```

```
ipaddress192.168.20.1255.255.255.0
```

```
noshutdown
```

```
exit end
```

Procedure

1. PlacetwoPCs, twoswitchesand onerouter inPacketTracer.
2. ConnectPC0 to Switch0.
3. ConnectSwitch0toRouterG0/0.
4. ConnectRouterG0/1to Switch1.
5. ConnectSwitch1 to PC1.
6. ConfiguretheIPaddresses.
7. Configurethedefaultgateways.
8. Configuretherouter interfaces.
9. UsethepingcommandfromPC0.

```
ping 192.168.20.10
```

ExpectedResult

```
Replyfrom192.168.20.10
```

```
Replyfrom192.168.20.10
```

```
Replyfrom192.168.20.10
```

```
Replyfrom192.168.20.10
```

Therefore,end-to-endcommunicationis successful.

WiresharkICMPAnalysis

UsethefollowingWiresharkfilter: icmp

The ICMP packet contains IPv4 header information such as:

Field	Meaning
Source IP Address	IP address of sender
Destination IP Address	IP address of receiver
TTL	Limit of the lifetime of the packet
Protocol	Identifies the upper-layer protocol
Header Length	Size of the IPv4 header

For ICMP, the protocol number is:

1

Role of Network Layer

The Network layer is responsible for:

- Logical IP Addressing
- Packet forwarding
- Routing between different networks
- Selecting the appropriate path
- Decreasing TTL at each router hop

Result

Communication between the two different LANs was successfully established through the router. ICMP packets were captured and IPv4 header fields were analyzed using Wireshark.

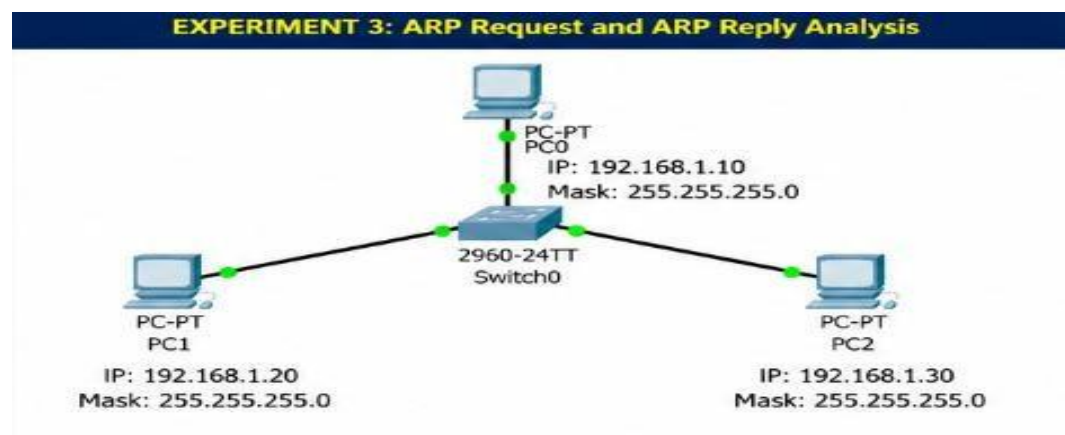
Experiment 3: ARP Request and ARP Reply Analysis Aim

To construct a LAN with multiple hosts, clear the ARP cache, initiate communication using ping, and analyze ARP Request and ARP Reply packets using Wireshark.

Requirements

- Cisco Packet Tracer
- Wireshark
- 3 PCs
- 1 switch
- Ethernet cables

NetworkTopology



IPAddressConfiguration

Device	IPAddress	SubnetMask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.20	255.255.255.0
PC2	192.168.1.30	255.255.255.0

Procedure

1. OpenCiscoPacketTracer.
2. AddthreePCsandoneswitch.
3. Connect all PCsto theswitch.
4. ConfiguretheIPaddresses.
5. CleartheARPcacheof thesourcePC.
6. StartWiresharkpacketcapture.
7. FromPC0,pingPC1:

ping 192.168.1.20

8. Observethe capturedpackets.
9. IdentifytheARPRequest.
10. IdentifytheARPReply.

ARPRequest

PC0initiallyknowsPC1'sIPaddressbutdoesnotknowitsMACaddress.

Therefore, PC0 broadcasts an ARP Request:

Who has 192.168.1.20?

Tell 192.168.1.10

The destination MAC address for the ARP Request is:

FF:FF:FF:FF:FF:FF

This is the Ethernet broadcast address.

ARP Reply

PC1 receives the ARP Request and sends an ARP Reply containing its MAC address. Conceptually:

192.168.1.20 is at XX:XX:XX:XX:XX:XX

The source PC then stores the IP-to-MAC mapping in its ARP cache.

ARP Address Mapping

IP Address



ARP Request



MAC Address



Ethernet Communication

ARP Packet Analysis

Field	ARP Request	ARP Reply
Sender IP	192.168.1.10	192.168.1.20
Target IP	192.168.1.20	192.168.1.10
Sender MAC	PC0 MAC	PC1 MAC
Target MAC	Unknown	PC0 MAC

How ARP Works

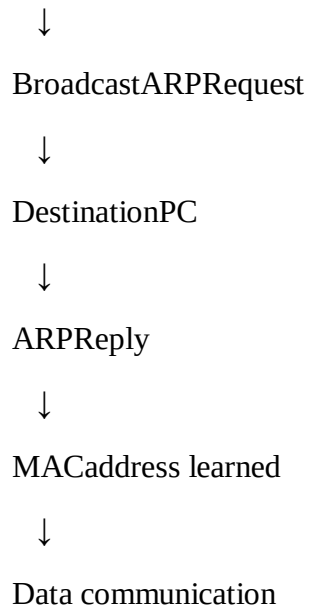
ARP converts a logical IPv4 address into the physical MAC address required for communication on the local Ethernet network.

The process is:

Source PC



Does not know destination MAC



Result

TheARP Request andARP Reply were successfully identified.ARP was used to resolve the destination IP address into its corresponding MAC address before normal Ethernet communication.

Experiment5:HTTPCommunicationandTCPThree-WayHandshakeAim

To create a client-server network using Cisco Packet Tracer, configure an HTTP server andclient PC, access a webpage, and analyze TCP and HTTP communication using Wireshark.

Requirements

- CiscoPacketTracer
- Wireshark
- 1 Server
- 1 PC
- 1 Switch
- Ethernetcables

NetworkTopology

EXPERIMENT 5: HTTP Communication and TCP Three-Way Handshake



IPAddressConfiguration

Device	IPAddress	SubnetMask
PC0	192.168.1.10	255.255.255.0
Server	192.168.1.20	255.255.255.0

ServerConfiguration

1. Open the Server.
2. Select **Desktop** → **IP Configuration**.
3. Set:

IPAddress: 192.168.1.20

SubnetMask: 255.255.255.0

4. Open **Services**.
5. Select **HTTP**.
6. Turn **HTTP service ON**.

Client

Configuration Configure PC0:

IPAddress: 192.168.1.10

SubnetMask: 255.255.255.0

Access the Webpage

Open the PC web browser and enter:

http://192.168.1.20

The server webpages should be displayed.

TCP Three-Way Handshake

Before HTTP data is transferred, TCP establishes a connection using three steps.

Step1– SYN

Client sends:

SYN

to the server.

Step2–SYN-ACK

Server responds:

SYN+ACK

Step3 –ACK

Client sends:

ACK

Therefore:

Client	Server
----- SYN ----->	
<-----SYN-ACK -----	
----- ACK----->	
TCP connection	
established	

HTTP GET Request

After the TCP connection is established, the client sends an HTTP GET request to obtain the webpage.

Conceptually:

GET/ HTTP/1.1

The server processes the request and sends an HTTP response.

HTTP Response

The server returns the requested webpage through an HTTP response. The communication sequence is:

TCP SYN

↓
TCPSYN-ACK
↓
TCPACK
↓
HTTPGET
↓
HTTPResponse

WiresharkAnalysis

UsefulWiresharkfiltersinclude:

tcp
and
http

The captured communication should show the TCP connection establishment followed by HTTP application-layer traffic.

HowTCPProvidesReliableDelivery

TCPprovidesreliablecommunication through:

1. **Connection** **establishment**
TCPestablishesaconnectionbeforetransferringapplication data.
2. **Acknowledgments**
Thereceiveracknowledgessuccessfullyreceived data.
3. **Sequence** **numbers**
TCPusessequencenumberstomaintainthecorrectorderof data.
4. **Retransmission**
Lost datacan beretransmitted.
5. **Ordered** **delivery**
Dataisdeliveredtotheapplicationinthecorrectsequence.

ApplicationLayer

HTTPworks at theApplication layer and is responsible for transferring web content between the client and server.

TCP works below HTTP and provides reliable transport.

Result

The client successfully accessed the webpage hosted by the server. The TCP three-way handshake, HTTP GET request, and HTTP response were identified. TCP's connection establishment, acknowledgment mechanism, and ordered data transfer provide reliable delivery of HTTP traffic.